

Data Types and Variables

Computing Fundamentals for Social Sciences

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Outline

- 1 Types of Data
- 2 Types of Variables
- 3 Practice and Summary

What is Data?

- Raw facts or measurements.
- Can be classified by how it is collected or organized.
- Correct classification helps choose the right analysis method.

Cross-Sectional Data

- Data collected at a single point in time.
- Example: Survey of income levels of 100 households in 2023.

R example:

```
# Cross-sectional data: incomes of 5 households
income <- c(45000, 52000, 48000, 50000, 47000)
mean(income)
summary(income)
```

Cross-Sectional Data: Visualizations

- Basic plots in R for cross-sectional data.

```
barplot(income, main="Income of 5 households",  
        xlab="Household", ylab="Income", col="skyblue")  
boxplot(income, main="Income Distribution", col="lightgreen"  
        )
```

Time-Series Data

- Data collected over time at regular intervals.

R example:

```
# Time-series: monthly temperature
temp <- c(15.2, 16.5, 18.1, 20.3, 22.0)
time <- 1:5 # e.g. months
plot(time, temp, type="o", main="Monthly Temperature",
      xlab="Month", ylab="Temperature")
```

Time-Series: Multiple Plots

```
# Adding trend line
plot(time, temp, type="o", main="Monthly Temperature", col="
      blue")
lines(time, temp, type="b", col="red")
```

Longitudinal (Panel) Data

- Observing multiple subjects over multiple time periods.

R example:

```
# Longitudinal data: 3 people over 4 years
income <- matrix(c(45,47,50,52,
                  40,42,45,48,
                  55,56,58,60), nrow=4, byrow=TRUE)

colnames(income) <- c("Person1", "Person2", "Person3")
rownames(income) <- paste("Year", 1:4)
income

# Plot each person's income over time
matplot(income, type="o", pch=1:3, col=1:3,
        xlab="Year", ylab="Income", main="Panel Data")
legend("topleft", legend=colnames(income), col=1:3, pch=1:3)
```


Spatial Data

- Data associated with geographical locations.
- Example: Pollution levels in different city locations.

R example:

```
locations <- data.frame(  
  lon = c(-73.95, -73.98, -74.01),  
  lat = c(40.75, 40.76, 40.74),  
  pollution = c(30, 45, 25)  
)  
plot(locations$lon, locations$lat, col=rainbow(3), pch=19,  
      cex=2)  
text(locations$lon, locations$lat, labels=locations$  
      pollution, pos=3)
```

Qualitative (Categorical) Variables

- Describe categories or groups.
- Nominal (no order) or ordinal (ordered).
- Examples: Gender, color, type of car.

Qualitative Variables: R Examples

```
gender <- factor(c("Male", "Female", "Female", "Male"))  
table(gender)  
  
# Bar plot  
barplot(table(gender), col=c("lightblue", "pink"))
```

Quantitative Variables

- Measured numerically.
- Can be discrete or continuous.

Discrete Variables

- Countable values, often integers.
- Example: Number of children in a family.

```
children <- c(0,2,3,1,4)
mean(children)
table(children)
barplot(table(children), col="lightgreen")
```

Continuous Variables

- Can take any value within an interval.
- Example: Height, weight, temperature.

```
height <- c(160.5,170.2,165.3,180.4)
summary(height)
hist(height, col="lightblue", main="Height Distribution",
      xlab="Height")
boxplot(height, col="lightgreen", main="Boxplot of Height")
```

Summary

- Data types: cross-sectional, time-series, longitudinal, spatial.
- Variables: qualitative (nominal/ordinal) or quantitative (discrete/continuous).
- Visualizations in R help understand distributions and patterns.

Practice Exercises

- 1 Create a cross-sectional vector and compute mean and summary.
- 2 Generate a simple time-series and plot it.
- 3 Create a longitudinal matrix and plot it with `matplot()`.
- 4 Define a qualitative factor and create a barplot.
- 5 Generate a histogram and boxplot for a continuous variable.

Exercise Solutions

```
# 1.
data <- c(10,15,20,25)
mean(data)
summary(data)

# 2.
ts_data <- ts(c(3,5,7,6,8), start=2020)
plot(ts_data, type="o")

# 3.
long_data <- matrix(c(1,2,3,4,5,6), nrow=3)
matplot(long_data, type="o")

# 4.
colors <- factor(c("red","blue","red","green"))
table(colors)
barplot(table(colors), col=rainbow(4))

# 5.
weights <- c(60,65,70,75)
hist(weights, col="lightblue")
boxplot(weights, col="lightgreen")
```